

Multiple Choice

1. **Answer:** A
Options: A) Sunday; B) Monday; C) Tuesday; D) Wednesday
Note: Correct option: A - Sunday
2. **Answer:** B
Options: A) None; B) One; C) Two; D) Three
Note: Correct option: B - One
3. **Answer:** D
Options: A) 0; B) 1; C) 11; D) 22
Note: Correct option: D - 22
4. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
5. **Answer:** D
Options: A) 0; B) -2; C) $a-2$; D) $(2+a)^{-1}$
Note: Correct option: D - $(2+a)^{-1}$

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'True, False'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '0'.

Long Form Response

11. **Answer:** Since $\cos B = \frac{3}{5}$, we have $\cos B = \frac{AB}{BC} = \frac{9}{BC} = \frac{3}{5}$. Then, we can see that we have $9 = BC \cdot \frac{3}{5}$, so $BC = 9 \cdot \frac{5}{3} = 15$. From the Pythagorean Theorem, we have $AC = \sqrt{BC^2 - AB^2} = \sqrt{15^2 - 9^2} = \sqrt{144} = 12$. Finally, we can find $\cos C$: $\cos C = \frac{AC}{BC} = \frac{12}{15} = \frac{4}{5}$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** When $x < 1$ or $x > 5$, $-x^2 + bx - 5 < 0$. That means that $-x^2 + bx - 5 = 0$ at $x = 1$ and $x = 5$. So, the parabola has roots at 1 and 5, giving us $(x - 1)(x - 5) = 0$. However, we also know the parabola opens downwards since the coefficient of x^2 is negative, so we have to negate one of the factors. We can now write $-x^2 + bx - 5 = (1 - x)(x - 5) = -x^2 + 6x - 5$. Thus, $b = 6$.

Note: Students should show all steps in their mathematical solution.